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Experiment-04

Study of Half-Wave and Full-Wave Rectifier

CSE251 - Electronic Devices and Circuits Lab

Objective

1. To build a half-wave rectifier circuit and understand its operating principle.
2. To build a full-wave rectifier circuit and understand its operating principle.

Equipments

1. p-n junction diode (1N4007) x 4
2. Resistor (10kΩ) - ×1
3. Capacitors (1μF, 4.7μF) - 1 each
4. Function Generator
5. Oscilloscope
6. Multimeter
7. Breadboard and Wires

Background Theory

Diodes are used to build rectifier circuits which convert the input sinusoidal voltage V_s to a unipolar output V_o. There are two types of rectifier circuits: (i) Half-wave rectifier and (ii) Full-wave rectifier.

Half-Wave (HW) Rectifier

The circuit of a half-wave rectifier is shown in the following figure:

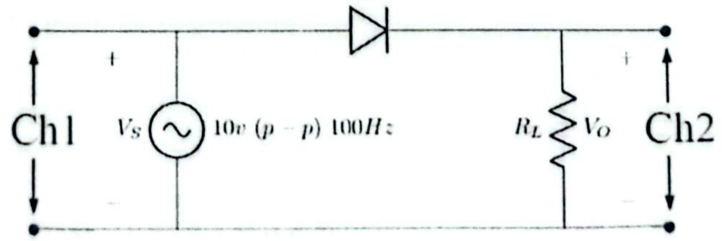


Figure 1: ***Half-Wave Rectifier***

Assuming ideal diode model,

For the period, $t = 0 \rightarrow T/2$, $V_S > 0$, Diode is ON and $V_O = V_S$

For the period, $t = T/2 \rightarrow T$, $V_S < 0$, Diode is OFF and $V_O = 0$

As only positive half cycle appears at the output and the negative half is blocked, the AC input voltage changes into a unidirectional DC voltage at the output. The process of removing half of the input signal to establish a dc level is aptly called half-wave rectification. Due to diode voltage drop, the actual output voltage will be

Data Sheet

Experimental Observation: HW Rectifier

inp max = 5.04

1. **HW Rectifier without Capacitor:**
   Diode voltage, V_{D0} = Input_{max} (oscilloscope) – Output_{max} (oscilloscope) = 0.56
   Maximum output voltage, V_p (oscilloscope) = 4.48V
   Average or DC output voltage, V_{dc} (multimeter in DC mode) = 1.368
   RMS or AC output voltage, V_{r-rms} (multimeter in AC mode) = 1.725

2. **HW Rectifier with 1μF Capacitor:**
   Diode voltage, V_{D0} = Input_{max} (oscilloscope) – Output_{max} (oscilloscope) = 0.48
   Maximum output voltage, V_p (oscilloscope) = 4.56
   Peak to peak ripple voltage, $V_{r(p-p)}$ (oscilloscope) = 2.64
   Average or DC value of the ripple voltage, V_{dc} (multimeter in DC mode) = 3.173
   RMS or AC value of the ripple voltage, V_{r-rms} (multimeter in AC mode) = 0.740
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.233

3. **HW Rectifier with 4.7μF Capacitor:**
   Diode voltage, V_{D0} = Input_{max} (oscilloscope) – Output_{max} (oscilloscope) = 0.72
   Maximum output voltage, V_p (oscilloscope) = 4.32
   Peak to peak ripple voltage, $V_{r(p-p)}$ (oscilloscope) = 0.88
   Average or DC value of the ripple voltage, V_{dc} (multimeter in DC mode) = 3.97
   RMS or AC value of the ripple voltage, V_{r-rms} (multimeter in AC mode) = 0.216
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.054

Theoretical Calculation: HW Rectifier

1. **HW Rectifier Without Capacitor:**
   Maximum output voltage, V_p (see the experimental observation) = 4.48V
   Maximum input voltage, V_m = 5.04V
   Diode voltage, V_{D0} (see the experimental observation) = 0.56V
   DC output voltage of the rectifier, $V_{dc} = \frac{V_m}{\pi} - \frac{V_{D0}}{2}$ = 1.32V
   RMS or AC output voltage, $V_{r-rms} = \frac{V_p}{2}$ = 2.24V

2. **HW Rectifier With 1μF Capacitor:**
   Maximum output voltage, V_p (see the experimental observation) = 4.56
   Peak to peak ripple voltage, $V_{r(p-p)}$ (see the experimental observation) = 2.64
   DC value of the ripple voltage, $V_{dc} = V_p - \frac{V_{r(p-p)}}{2}$ = 3.24
   RMS value of the ripple voltage, $V_{r-rms} = \frac{V_{r(p-p)}}{2\sqrt{3}}$ = 0.762
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.235

3. **HW Rectifier with 4.7μF Capacitor:**
   Maximum output voltage, V_p (see the experimental observation) = 4.32
   Peak to peak ripple voltage, $V_{r(p-p)}$ (see the experimental observation) = 0.88
   DC value of the ripple voltage, $V_{dc} = V_p - \frac{V_{r(p-p)}}{2}$ = 3.88
   RMS value of the ripple voltage, $V_{r-rms} = \frac{V_{r(p-p)}}{2\sqrt{3}}$ = 0.254
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.0654

Experimental Observation: FW Rectifier

1. **FW Rectifier without Capacitor:**
   Diode voltage, $V_{D0} = \frac{\text{Input}_{max}\text{(oscilloscope)} - \text{Output}_{max}\text{(oscilloscope)}}{2}$ = 0.56
   Maximum output voltage, V_p (oscilloscope) = 3.92
   Average or DC output voltage, V_{dc} (multimeter in DC mode) = 2.224
   RMS or AC output voltage, V_{r-rms} (multimeter in AC mode) = 1.341

2. **FW Rectifier with 1μF Capacitor:**
   Diode voltage, $V_{D0} = \frac{\text{Input}_{max}\text{(oscilloscope)} - \text{Output}_{max}\text{(oscilloscope)}}{2}$ = 0.55
   Maximum output voltage, V_p (oscilloscope) = 3.94
   Peak to peak ripple voltage, $V_{r(p-p)}$ (oscilloscope) = 1.22
   Average or DC value of the ripple voltage, V_{dc} (multimeter in DC mode) = 3.325
   RMS or AC value of the ripple voltage, V_{r-rms} (multimeter in AC mode) = 0.348
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.104

3. **FW Rectifier with 4.7μF Capacitor:**
   Diode voltage, $V_{D0} = \frac{\text{Input}_{max}\text{(oscilloscope)} - \text{Output}_{max}\text{(oscilloscope)}}{2}$ = 0.61
   Maximum output voltage, V_p (oscilloscope) = 3.82
   Peak to peak ripple voltage, $V_{r(p-p)}$ (oscilloscope) = 0.41
   Average or DC value of the ripple voltage, V_{dc} (multimeter in DC mode) = 3.591
   RMS or AC value of the ripple voltage, V_{r-rms} (multimeter in AC mode) = 0.091
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.025

Theoretical Calculation: FW Rectifier

1. **FW Rectifier without Capacitor:**
   Maximum output voltage, V_p (see the experimental observation) = 3.92
   Maximum input voltage, V_m = 5.04
   Diode voltage, V_{D0} (see the experimental observation) = 0.56
   DC output voltage of the rectifier, $V_{dc} = \frac{2\,V_m}{\pi} - 2\,V_{D0}$ = 2.088
   RMS or AC output voltage, $V_{r-rms} = \frac{V_p}{\sqrt{2}}$ = 2.77

2. **FW Rectifier with 1μF Capacitor:**
   Maximum output voltage, V_p (see the experimental observation) = 3.94
   Peak to peak ripple voltage, $V_{r(p-p)}$ (see the experimental observation) = 1.22
   DC value of the ripple voltage, $V_{dc} = V_p - \frac{V_{r(p-p)}}{2}$ = 3.33
   RMS value of the ripple voltage, $V_{r-rms} = \frac{V_{r(p-p)}}{2\sqrt{3}}$ = 0.3521
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.106

3. **FW Rectifier with 4.7μF Capacitor:**
   Maximum output voltage, V_p (see the experimental observation) = 3.82
   Peak to peak ripple voltage, V_p (see the experimental observation) = 0.4
   DC value of the ripple voltage, $V_{dc} = V_p - \frac{V_{r(p-p)}}{2}$ = 3.62
   RMS value of the ripple voltage, $V_{r-rms} = \frac{V_{r(p-p)}}{2\sqrt{3}}$ = 0.11
   Ripple factor, $r = V_{r-rms}/V_{dc}$ = 0.032

Table for Comparison

		Experimental Observation			Theoretical Calculation		
	C(μF)	V_{r-rms} (V)	V_{dc} (V)	Ripple Factor, r	V_{r-rms} (V)	V_{dc} (V)	Ripple Factor, r
HW	1	0.74	3.173	0.233	0.762	3.24	0.235
	4.7	0.216	3.97	0.054	0.254	3.88	0.0654
FW	1	0.348	3.325	0.104	0.3521	3.33	0.106
	4.7	0.091	3.591	0.025	0.11	3.62	0.032

3-12-2025
Signature of the lab faculty

Report Submission Guidelines

1. Attach the signed Data Sheet (if any)
2. Attach the captured images (if any)
3. Answer the questions in the "Test Your Understanding" section
4. Add a brief Discussion regarding the experiment. For the Discussion part of the lab report, you should include the answers of the following questions in your own words:
   - What did you learn from this experiment?
   - What challenges did you face and how did you overcome the challenges? (if any)
   - What mistakes did you make and how did you correct the mistakes? (if any)
   - How will this experiment help you in future experiments of this course?

Test Your Understanding

Answer the following questions:

1. Which capacitor acts as a better filter? Explain briefly.
   Answer:

   4.7uF, it has a higher capacitance so more charge is stored so there is less ripple. and the voltage output remains more smooth. and the output is more stable,

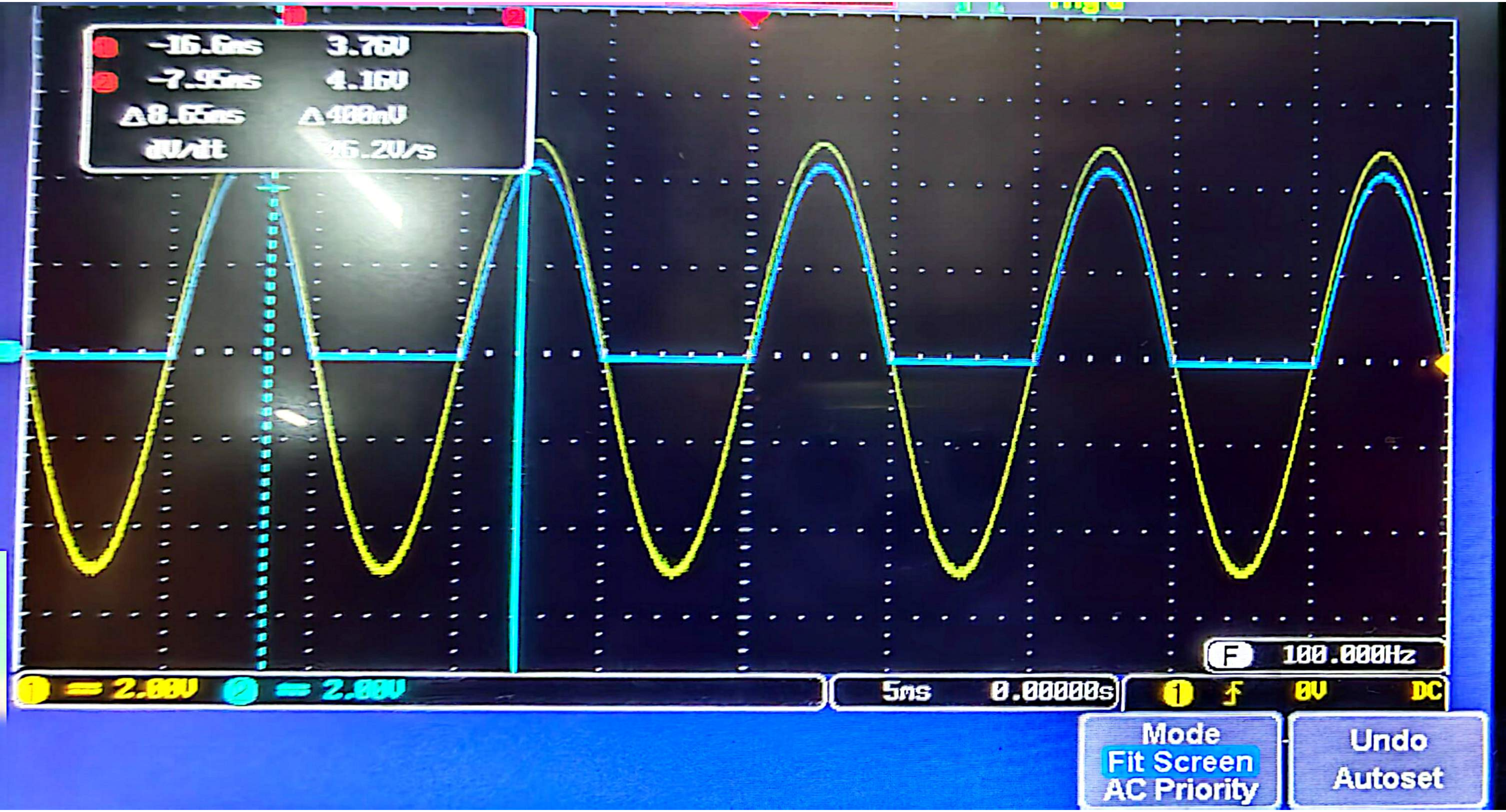
2. Which of the two rectifiers is better? Which quantity is calculated for this purpose? Explain briefly.
   Answer:

   Full wave bridge rectifier is better as full wave gives two pulses per cycle, while half wave gives one.
   More pulses give smoother output and theres less ripple.
   We calculate ripple factor to know which rectifier is better,
   we know $\gamma = \frac{V_{rms}}{V_{DC}}$, the higher the V_{DC}, the lower the γ the better the rectifier as less ripple, Full wave bridge has a lower γ.

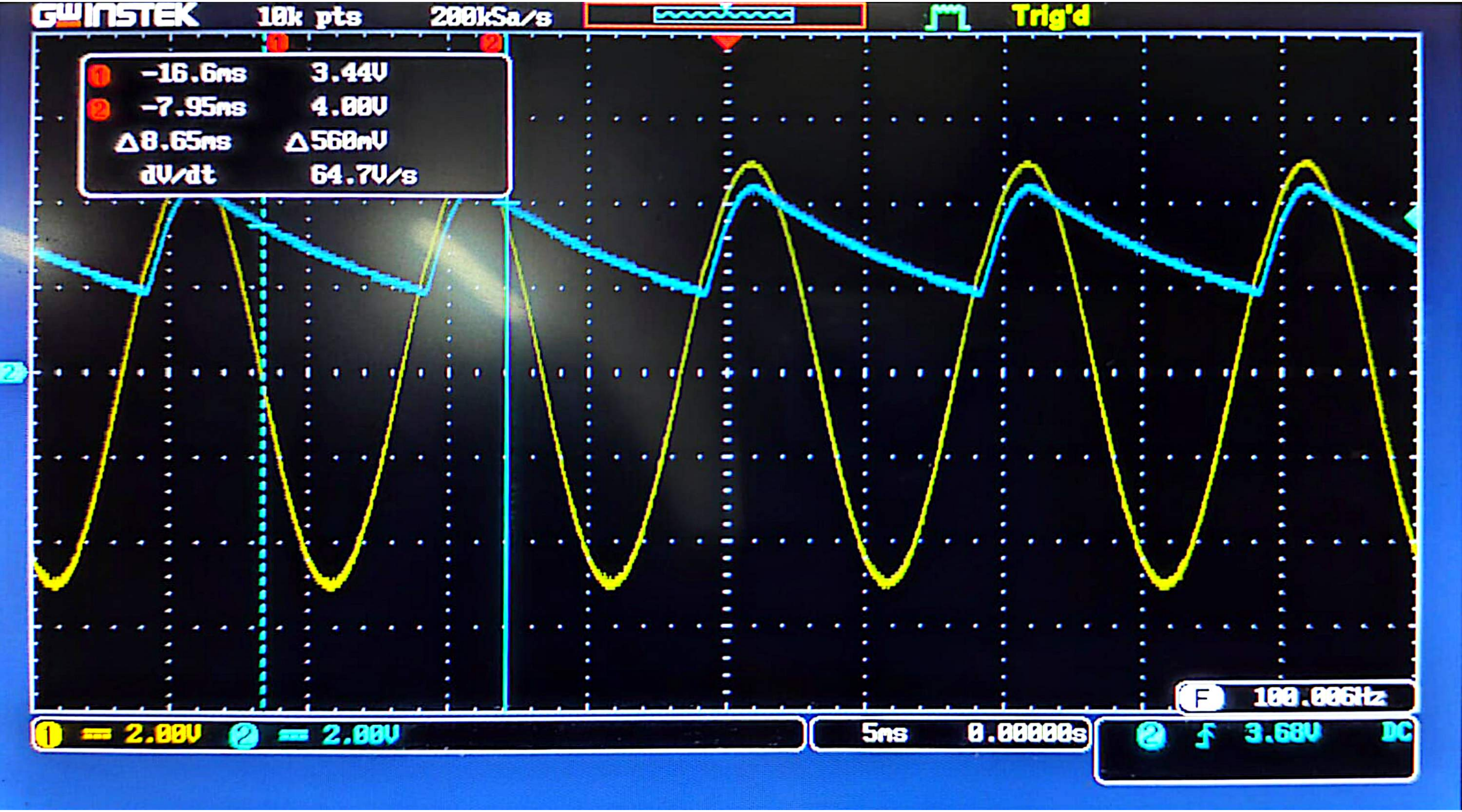
3. Why can't you see the input and output using both channels of the oscilloscope simultaneously in Task-02?
Answer:

Both channels have their grounds internally shorted.
In full bridge rectifiers, the AC input has no ground reference and the DC output has its own ground after the diodes.
If both channels are connected, AC and DC reference points get shorted and one waveform would become noisy or disappear, as there is a ground conflict.

I learned about rectifiers in this experiment, mainly how to setup a full bridge rectifier. The instructions given for this experiment were clear and easy to understand, so I didnt face much of a challenge. The only mistake we made was we forgot to take a few measurements because we were going too fast so we had to redo the experiment to take those measurements. This experiment may help me construct a power supply from AC power source.



-16.6ms 3.76V
-7.95ms 4.16V
Δ8.65ms Δ400mV
dV/dt 46.2V/s
F 100.000Hz
1 = 2.00V 2 = 2.00V
5ms 0.00000s
1 0V DC
Mode
Fit Screen
AC Priority
Undo
Autoset


GW INSTEK
10k pts
200kSa/s
Trig'd
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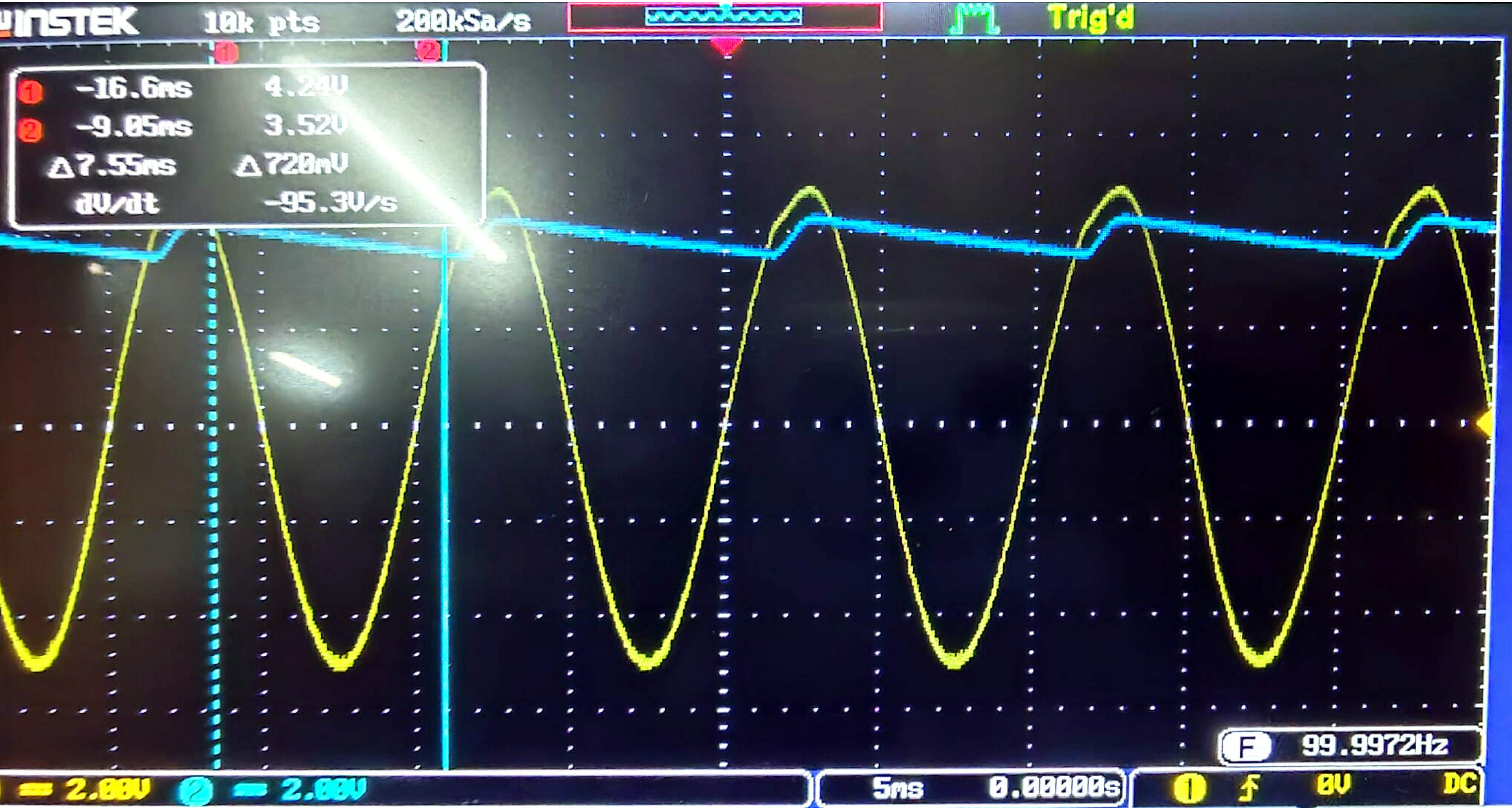
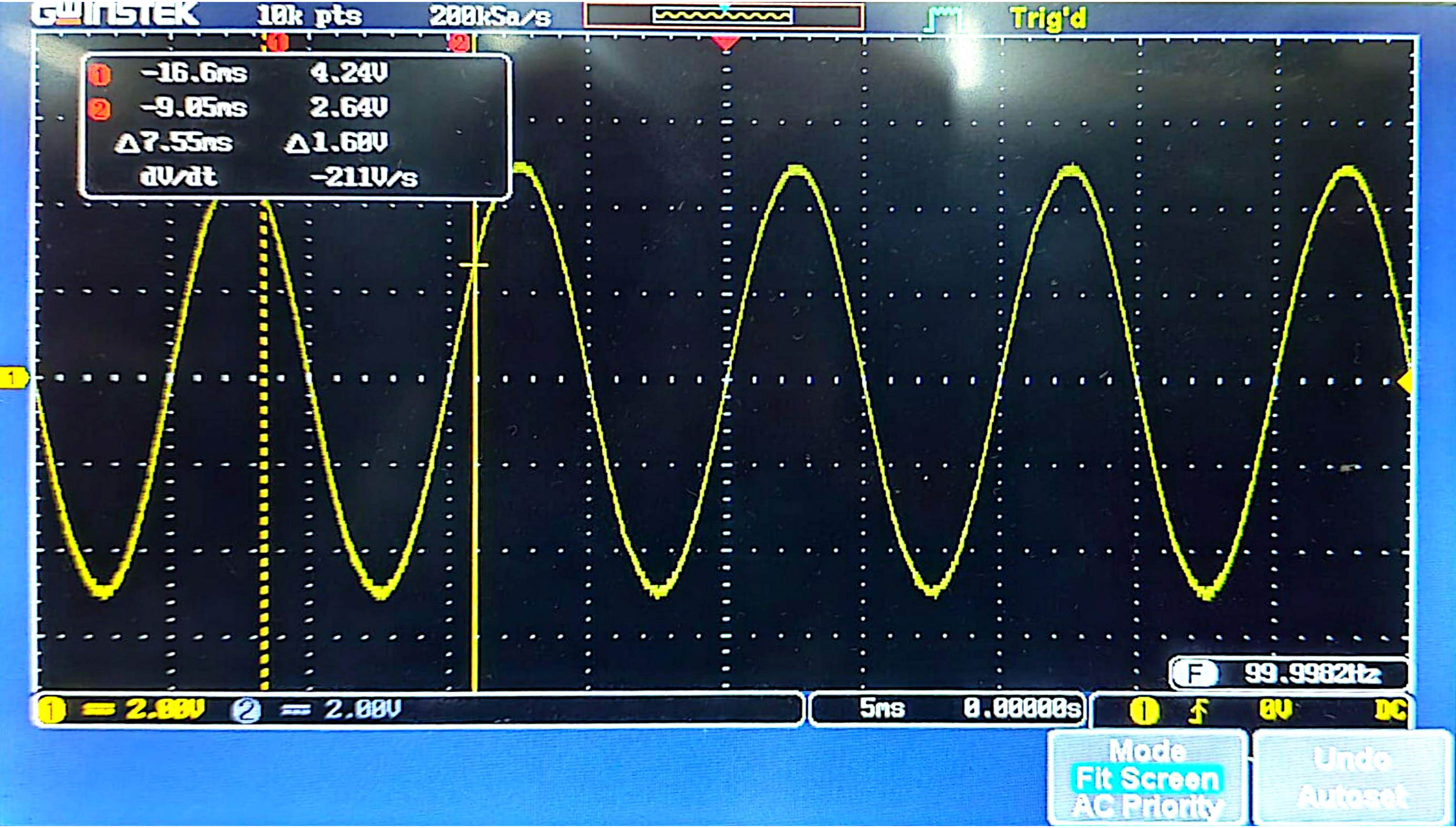
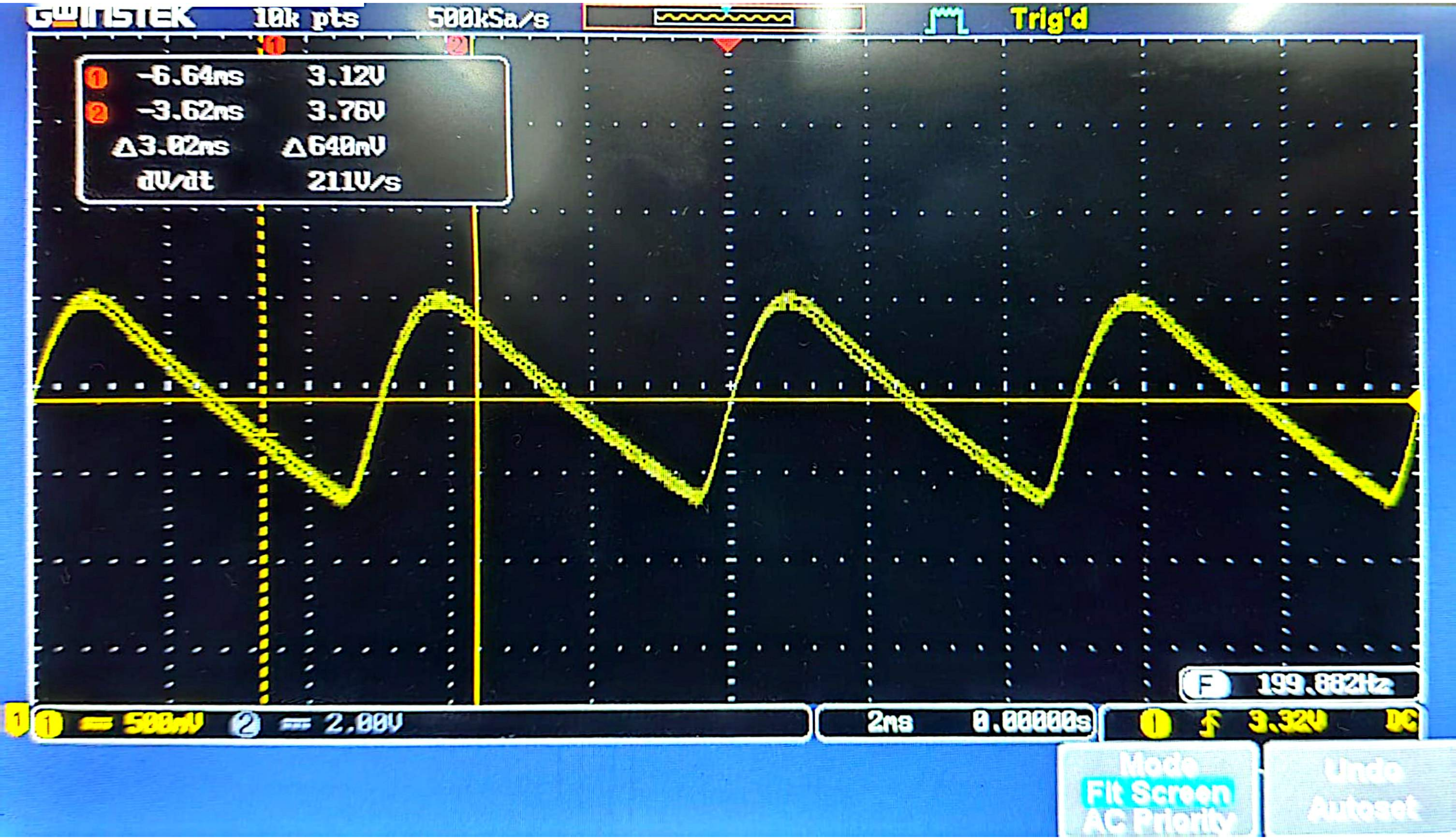
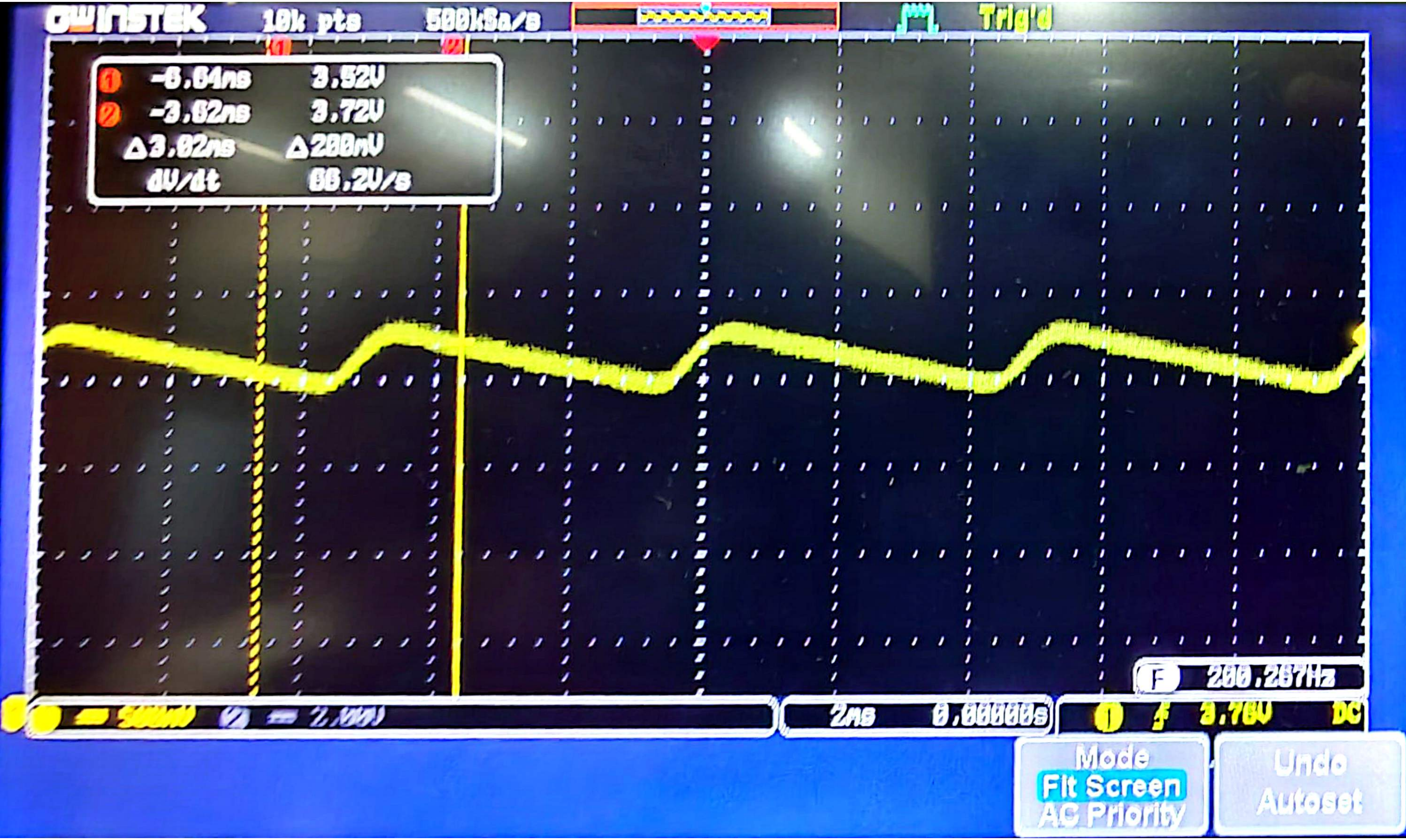
2 -7.95ms 4.00V
Δ8.65ms Δ560mV
dV/dt 64.7V/s
F 100.006Hz
1 2.00V
2 2.00V
5ms 0.00000s
2 3.68V DC


INSTEK
10k pts
200kSa/s
Trig'd
-16.6ms 4.24V
-9.05ms 3.52V
Δ7.55ms Δ720mV
dV/dt -95.3V/s
F 99.9972Hz
2.00V
2.00V
5ms 0.00000s
0V
DC


GWINSTEK
10k pts
200kSa/s
Trig'd
-16.6ms 4.24V
-9.05ms 2.64V
Δ7.55ms Δ1.60V
dV/dt -211V/s
99.9982Hz
1 = 2.00V
2 = 2.00V
5ms 0.00000s
0V
DC
Mode
Fit Screen
AC Priority
Undo
Autoset


GW INSTEK
10k pts
500kSa/s
Trig'd
1 -6.64ms 3.12V
2 -3.62ms 3.76V
Δ3.02ms Δ640mV
dV/dt 211V/s
F 199.882Hz
1 500mV
2 2.00V
2ms 0.00000s
1 3.32V DC
Mode
Fit Screen
AC Priority
Undo
Autoset


GW INSTEK
10k pts
500kSa/s
Trig'd
-6.84ns 3.52V
-3.82ns 3.72V
Δ3.02ns Δ200mV
dV/dt 66.2V/s
200.267Hz
500mV
2.00V
2ms
0.00000s
3.78V
DC
Mode
Fit Screen
AC Priority
Undo
Autoset

GWINSTEK
10k pts
500kSa/s
-6.64ms 1.00V
-3.62ms 3.88V
Δ3.02ms Δ2.88V
dV/dt 953V/s
F 200.088Hz
1.00V
2.00V
2ms 0.00000s
1.96V
DC